MATH 1110
Calculus 1: Quiz 1
Given Time: 1 hr.

Name: _____ Student ID: _____

INSTRUCTION: Show all the necessary steps on the provided papers

1. Which of the following is a simplified form of $2(3m + x) + 5x$.

$A.$ $6m + 6x$ $C.$ $6m + 8x$

$B.$ $6m + 5x$ $D.$ $6m + 7x$

2. If we expand the expression $(4x - 5)^2$, we get

$A.$ $16x^2 - 40x + 25$ $C.$ $16x^2 + 25$

$B.$ $16x^2 - 40x - 25$ $D.$ $16x^2 - 25$

3. The solution set of $|4x - 3| = 1$.

$A.$ $\{1, -\frac{1}{3}\}$ $C.$ $\{1, \frac{1}{2}\}$

$B.$ $\{-1, \frac{1}{2}\}$ $D.$ $\{\}$

4. Which of the following is the solution of $|2x - 1| < 5$

$A.$ $[-2, 3]$ $C.$ $(-3, 2)$

$B.$ $(-2, 3)$ $D.$ $(2, 3)$

5. Let $f(x) = \frac{x+1}{x^2-4}$. Which of the following is false.

$A.$ $f(x)$ is a rational function

$B.$ $y = 0$ is the horizontal asymptote

$C.$ The domain is all real numbers except ± 2

$D.$ $x = -1$ is the vertical asymptote

6. Let $f(x) = x^2 - 2$. Which of the following is equal to $\frac{f(x)-f(3)}{x-3}$.

A. $x + 3$

B. $\frac{1}{x+3}$

C. $\frac{1}{x-3}$

D. $\frac{x+3}{x-3}$

7. Let $f(x) = 3x^2 - 1$ and $g(x) = 5x$. Which of the following is true?

A. $(f \circ g)(x) = 15x^2 - 1$

B. $(g \circ g)(x) = 25x^2$

C. $(g \circ f)(x) = 15x^2 - 5$

D. $(f \circ g)(x) = 75x^2 - 5$

8. Let $f(x) = x + \frac{1}{x}$ and $g(x) = \frac{x}{x-2}$. Which of the following is true?

A. $f(g(x)) = x - 2$

B. $f(f(x)) = x$

C. $g(f(x)) = \frac{x}{1-2x}$

D. None of the above.

9. Let $f(x) = 3x - 1$. The inverse $f^{-1}(x)$ is

A. $\frac{x+1}{3}$

B. $\frac{x-1}{3}$

C. $\frac{1}{3x-1}$

D. $\frac{3}{x} - 1$

10. Let $f(x) = \frac{3}{3x+4}$. The inverse of $f^{-1}(x)$ is

A. $\frac{3-4x}{3x}$

B. $\frac{4x+3}{3x}$

C. $\frac{3x-4}{3}$

D. $\frac{3x}{3-4x}$

11. Which of the following is the equation of the line in the $xy-$plane with slope -5 that contains the point $(1, -3)$

A. $y = -5x + 3$

B. $= -\frac{1}{5}x + 2$

C. $y = -5x - 2$

D. $y = -5x + 2$

12. If the point $(-4, t)$ is on the line containing the points $(1, 2)$ and $(0, 8)$, then $t =$

$A.\ -8$ $\quad B.\ 32$ $\quad C.\ -32$ $\quad D.\ -16$

13. The equation of the line in the $xy-$plane that contains the point $(4, 5)$ and that is parallel to the line $y - 2x + 1 = 0$ is

$A.\ 2x + y = 2$ $\quad B.\ x + 2y = 7$ $\quad C.\ 2x - y = 3$ $\quad D.\ y - 2x = 3$

14. Solve the quadratic equation $x^2 + 7x + 12 = 0$.

$A.\ \{-3, 4\}$ $\quad B.\ \{-3, -4\}$ $\quad C.\ \{3, -4\}$ $\quad D.\ \{3, 4\}$

15. The solution of the quadratic equation $x^2 - 6x - 2 = 0$.

$A.\ \{\ 3 - \sqrt{11},\ \ 3 + \sqrt{11}\ \}$

$B.\ \{\ -8, 14\ \}$

$C.\ \{\ 11 - \sqrt{3},\ \ 11 + \sqrt{3}\ \}$

$D.\ \{\ 6, 2\ \}$

16. Which if the following is the simplified form the expression

$$\left(\frac{\left(x^2y^{-1}\right)^3}{\left(x^3y^2\right)^2}\right)^2.$$

$A.\ y^{14}$ $\quad B.\ y^{-14}$ $\quad C.\ x^{12}$ $\quad D.\ x^{-12}$

17. Let $p(x) = 3x^2 + 5x + 2$, $q(x) = -3x^2 - 2x - 1$. Find $p(x) - q(x)$

$A.$ $6x^2 + 7x + 3$ $C.$ $3x + 1$

$B.$ $-6x^2 - 7x - -3$ $D.$ $7x + 3$

18. Let $f(x) = x^2 - 6x + 8$. Which of the following is true?

$A.$ the vertex is $(-3, -1)$ $C.$ $x = 2$ *and* $x = 4$ are the x-intercepts

$B.$ the minimum value of the function is 1 $D.$ the graph is a parabola opening down

19. Let $f(x) = \frac{2x-3}{x+2}$. Which of the following is false?

$A.$ $x = \frac{3}{2}$ is the x-intercept $C.$ $x = -2$ the vertical asymptote

$B.$ $y = \frac{-3}{2}$ is the y-intercept $D.$ $y = \frac{-3}{2}$ is horizontal asymptote

20. If $5^{-x} = \frac{1}{125}$, then $x =$.

$A.$ 3 $B.$ -3 $C.$ 125 $D.$ $\log 5$

21. If $2^{x-2} = 8^{2x+1}$, then $x =$.

$A.$ 1 $B.$ -1 $C.$ $-\frac{3}{5}$ $D.$ -3

22. Which of the following is true?

$A.$ $\log_4 64 = 4$ $B.$ $\log_8 2 = \frac{-1}{3}$ $C.$ $\log_8 2 = 3$ $D.$ $\log 0.001 = -3$

23. If $\log_2(x+4) = 3$, then $x =$

$A.$ 5 $B.$ 4 $C.$ -2 $D.$ -5

24. If $\log_3 x = 1.2$ and $\log_3 y = 1.3$, then $\log_3\left(\frac{9x^2}{y}\right) =$

$A.$ 2 $B.$ 10.1 $C.$ 3.1 $D.$ 1.1

25. Which of the following is true?

$A.$ $f(x) = 2^x$ is a decreasing function.

$B.$ $f(x) = 2^{-x}$ is an increasing function.

$C.$ $f(x) = \log_2 x$ is a decreasing function.

$D.$ $f(x) = \log_2 x$ is a increasing function.

26. Which of the following is false?

$A.$ $150^\circ = \frac{5\pi}{6}$ radian $C.$ $390^\circ = \frac{13\pi}{6}$ radian

$B.$ $135^\circ = \frac{3\pi}{4}$ radian $D.$ $60^\circ = \frac{\pi}{6}$ radian

27. The end-point (x, y) of the radius of the unit circle that corresponds to the angle $\theta = 5\pi/6$ is

$A.$ $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ $B.$ $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ $C.$ $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ $D.$ $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

28. If $\cos\theta = -\frac{4}{5}$, where $\pi < \theta < 3\pi/2$, then $\tan\theta =$

$A.$ $\frac{3}{4}$ $B.$ $\frac{4}{3}$ $C.$ $-\frac{3}{4}$ $D.$ $-\frac{4}{3}$

29. Suppose $0 < \theta < \pi/2$ and $\sin\theta = 1/3$.Which of the following is false?

$A.$ $\cos\theta = \frac{\sqrt{8}}{3}$ $B.$ $\tan\theta = 3$ $C.$ $\cot\theta = \sqrt{8}$ $D.$ $\sec\theta = \frac{3}{\sqrt{8}}$

30. In a right angle triangle ABC, suppose $C = 90°$, a is the side opposite angle A, b the side opposite angle B, and c is the hypothenuse. If $a = 2$ and $b = 3$. which of the following is ture?

$A.$ $\cos C = 1$ $B.$ $\tan B = \frac{3}{2}$ $C.$ $\sec A = \frac{\sqrt{13}}{2}$ $D.$ $\sin A = \frac{3}{\sqrt{13}}$

31. If $\cos\theta = -\frac{\sqrt{3}}{2}$ and $\sin\theta = \frac{1}{2}$, then $\sin(2\theta) =$

$A.$ $-\frac{\sqrt{3}}{2}$ $B.$ $-\frac{1}{2}$ $C.$ $\frac{\sqrt{3}}{2}$ $D.$ $\frac{1}{2}$

32. Which of the following is true?

$A.$ $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right) = \frac{2\pi}{3}$

$B.$ $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) = \frac{4\pi}{3}$

$C.$ $\sin\left(\sin^{-1}\left(\frac{1}{3}\right)\right) = \frac{1}{3}$

$D.$ $\tan\left(\tan^{-1}(3)\right) = \pi$